Cholesky.java

Below is the syntax highlighted version of Cholesky.java from §9.5 Numerical Solutions to Differential Equations.

```
/*************************************************************************
 *  Compilation:  javac Cholesky.java
 *  Execution:    java Cholesky
 *
 *  Compute Cholesky decomposition of symmetric positive definite
 *  matrix A = LL^T.
 *
 *  % java Cholesky
 *  2.00000  0.00000  0.00000
 *  0.50000  2.17945  0.00000
 *  0.50000  1.26179  3.62738
 *
 *************************************************************************/

public class Cholesky {
    private static final double EPSILON = 1e-10;

    // is symmetric
    public static boolean isSymmetric(double[][] A) {
        int N = A.length;
        for (int i = 0; i < N; i++) {
            for (int j = 0; j < i; j++) {
                if (A[i][j] != A[j][i]) return false;
            }
        }
        return true;
    }

    // is symmetric
    public static boolean isSquare(double[][] A) {
        int N = A.length;
        for (int i = 0; i < N; i++) {
            if (A[i].length != N) return false;
        }
        return true;
    }


    // return Cholesky factor L of psd matrix A = L L^T
    public static double[][] cholesky(double[][] A) {
        if (!isSquare(A)) {
            throw new RuntimeException("Matrix is not square");
        }
        if (!isSymmetric(A)) {
            throw new RuntimeException("Matrix is not symmetric");
        }

        int N  = A.length;
        double[][] L = new double[N][N];

        for (int i = 0; i < N; i++)  {
            for (int j = 0; j <= i; j++) {
                double sum = 0.0;
```

```
                for (int k = 0; k < j; k++) {
                    sum += L[i][k] * L[j][k];
                }
                if (i == j) L[i][i] = Math.sqrt(A[i][i] - sum);
                else        L[i][j] = 1.0 / L[j][j] * (A[i][j] - sum);
            }
            if (L[i][i] <= 0) {
                throw new RuntimeException("Matrix not positive definite");
            }
        }
        return L;
    }


    // sample client
    public static void main(String[] args) {
        int N = 3;
        double[][] A = { { 4, 1,  1 },
                         { 1, 5,  3 },
                         { 1, 3, 15 }
                       };
        double[][] L = cholesky(A);
        for (int i = 0; i < N; i++) {
            for (int j = 0; j < N; j++) {
                StdOut.printf("%8.5f ", L[i][j]);
            }
            StdOut.println();
        }

    }

}
```

Copyright © 2000–2011, Robert Sedgewick and Kevin Wayne.
Last updated: Tue Aug 30 09:58:33 EDT 2016.